AJAY CHOUDHARY SIR • CHEMISTRY ACADEMY

8+ Years Teaching Excellence • Organic | Inorganic | Practical Chemistry

Chemical Bonding & Molecular Structure Worksheet

VSEPR, Hybridization & Molecular Orbital Theory Drill

Chapter / Focus Area: Class 11 Foundation & JEE/NEET Target

1. Hybridization & Geometry Problems

• Calculate steric number for SF_4 (sp^3d, see-saw shape, 1 lone pair).

• Calculate steric number for XeF_4 (sp^3d^2, square planar shape, 2 lone pairs).

• Predict bond angles in H_2O (104.5°), NH_3 (107°), and CH_4 (109.5°) using VSEPR theory.

2. Molecular Orbital Theory (MOT) Applications

• Oxygen Molecule (O_2): Bond order = (10 - 6)/2 = 2. Paramagnetic due to 2 unpaired electrons in $\pi^*(2p_x)$ and $\pi^*(2p_y)$.

• Nitrogen Molecule (N_2): Bond order = (10 - 4)/2 = 3. Diamagnetic.